EOPTIC
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REQUEST FOR QUOTE

IMX541 MIPI ELECTRONICS DEVELOPMENT

Submitted To:

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Revision History

Rev	Date	Author	Description
A	2026 06 12	N. Cox	Initial release
B	2026 06 12	N. Cox	Req. Update
C	2026 06 29	N. Cox	Added Appendix B - Technical Requirements

Executive Summary

Eoptic is requesting a proposal/quote for development of camera electronics for a **SONY IMX541** image sensor to a **MIPI CSI-2 output** suitable for processing by an embedded FPGA and/or GPU image processing unit. The electronics shall be designed and qualified for the LEO space environment including radiation tolerance, thermal cycling, and launch vibration loads.

Key Requirements

01 **Design Deliverables Deposited in Escrow**
Kaya agrees to deposit all design deliverables as detailed in Appendix A.

02 **Bill of Materials with Country of Origin**
A complete Bill of Materials (BOM) identifying country of origin for every component is a mandatory deliverable. This requirement supports supply chain compliance and export control obligations.

03 **Dual Pricing Path: US-Domestic and International**
Proposals shall include separate unit and NRE pricing for: (1) a US-domestic supply chain path, and (2) an international supply chain path.

04 **Technical Implementation**
See **Appendix B** for full technical requirements. At minimum camera electronics need to enable:

- On Sensor 2x2 binning functionality
- Multiple read out modes: ROI/ partial frame readout, full resolution, full-FPS, selectable ADC bit-depth support etc…
- Noise floor non-degradation
- Field-upgradable firmware

Anticipated Production Quantities

Pricing shall be provided at each of the following production tiers. NRE shall be quoted separately from unit recurring cost.

Period	Unit Quantity	Notes
Delivery (no later than) Jan 2027	15*	Initial delivery / qualification build
Delivery Q2-Q3 2028	40-70	Low-rate initial production (LRIP)
5 years beginning Q2-Q3 2029	400-750 / yr	Full-rate production: Monthly or quarterly delivery cadence TBD

*Include in initial delivery QTY-1 GSE to sufficient to power, configure, and operate the delivered camera electronics and to demonstrate all specified sensor features (sensor configuration, image acquisition, display, documentation, etc..)

Appendix A

IP Escrow Agreement

Kaya agrees to deposit all design deliverables (including schematics, PCB layout files, FPGA/firmware source code, test procedures, and associated documentation) into a mutually agreed third-party escrow account administered by a recognized escrow agent (e.g., Iron Mountain Intellectual Property Management, or equivalent). The escrow deposit shall be maintained and updated at mutually agreed program milestones and upon each major design revision.

Release Conditions

The escrowed materials shall be released to Eoptic upon the occurrence of any of the following triggering events:

1. Insolvency or Bankruptcy — Kaya files for, or is placed into, bankruptcy, insolvency, receivership, liquidation, or any analogous proceeding under applicable law, and is unable to fulfill its production obligations.

2. Cessation of Business — Kaya ceases operations, winds down its business, or otherwise discontinues the production of the subject product(s) and provides no qualified successor or sublicensed manufacturer acceptable to Eoptic.

3. Ownership or Control Change to a Restricted Jurisdiction — Kaya undergoes a change of ownership, acquisition, merger, or effective control that results in majority ownership or operational control being held by an entity domiciled in, or subject to the laws of, a country with which Eoptic is legally or contractually prohibited from conducting business (including but not limited to countries subject to applicable export control regulations, sanctions, or government-mandated restrictions).

4. Failure to Supply — Kaya fails to meet its delivery obligations for a continuous period of 120 days and is unable to demonstrate a credible remediation plan within 30 days of written notice from Eoptic, with no mutually acceptable alternative production arrangement in place.

5. Material Breach — Kaya commits a material breach of the production or supply agreement governing this program and fails to cure such breach within the applicable cure period.

6. Voluntary Discontinuation of Product — Kaya elects to discontinue manufacturing the subject product and provides written notice of such discontinuation, with no qualified successor manufacturer arranged.

7. Force Majeure Extended Interruption — A force majeure event prevents Kaya from fulfilling production obligations for a continuous period exceeding 180 days with no reasonable prospect of resumption.

8. Export Control Disqualification — Kaya becomes subject to export control restrictions, disbarment, or sanctions that prevent it from lawfully supplying the product to E-optic.

Escrow Terms

- Costs of establishing and maintaining the escrow shall be the responsibility of E-optic.
- E-optic's right to access the escrowed materials upon a triggering event shall be limited to internal use for continued production of the subject product(s) and shall not convey broader IP ownership or rights to sublicense.
- KAYA retains all intellectual property rights in the escrowed materials at all times absent a separately negotiated ownership transfer.
- The parties shall execute a standalone IP Escrow Agreement prior to program kickoff, defining the escrow agent, deposit schedule, verification rights, and release procedures in detail.

Appendix B - Technical Requirements

General

The camera electronics shall have capability parity with commercially available products (e.g. Allied Vision Alvium C) and additionally address space-specific requirements.

- Eoptic to provide mechanical dimensions outline (~26.4 x 26.4mm)
- MIPI CSI-2
  - 4-Lane, 1.5Gbps/lane nominal
    - Min 21FPS at Raw12 Full Resolution
- HW Trigger input / Sync output
- Software configurable capabilities
  - Raw 8/10/12 bit modes
  - Readout modes:
    - All-pixels
    - 2 x 2 Floating Diffusion FD (on sensor) binning
    - ROI mode
    - Vert/Horix normal + inverted readout
- Image Quality Pipeline (Non-volatile memory stored maps/coefficients)
  - Defect pixel correction
  - Fixed pattern noise / NUC
  - LUT